

Lec 2

Signal Operations: Shifting, Folding & Scaling

Notes — 20th July 2026

1. Shifting Operation

Trigger: Whenever addition or subtraction is involved inside the argument of $x(t)$, e.g. $x(t + 1)$, $x(t - 1)$.

Key property: During shifting, the **shape and amplitude of the signal remain unchanged** — only its position on the time axis changes.

How to perform a shift

1. Draw the graph and label the axes.
2. Equate the bracket term to 0 to find the shift point.
3. Solve for t .

“Pasted image 20260720152807.png” could not be found.

Example — $x(t+1)$

$$\begin{aligned}t + 1 &= 0 \\t &= -1\end{aligned}$$

“Pasted image 20260720152949.png” could not be found.

“Pasted image 20260720153018.png” could not be found.

- The **negative sign** on t indicates a **left shift** (important).
- The magnitude (1) indicates the **unit** of shift.
- Shape and amplitude remain the same.

Time Delay vs Time Advance

Direction	Name	Sign convention
Shift right	Time Delay	Corresponds to $t - a$ (positive shift)
Shift left	Time Advance	Corresponds to $t + a$ (negative shift)

Intuition: A negative sign associated with the shift simply represents a **change in direction** (left vs right), not a change in shape/amplitude.

2. Folding Operation

“Pasted image 20260720154440.png” could not be found.

Trigger: Whenever t itself becomes **negative** inside the argument — i.e., $x(-t)$.

Key property: Folding produces a **mirror image of the signal about the y-axis** (symmetric reflection).

Rule of thumb

- $x(-kt) \rightarrow$ **Folding operation**
- $x(kt) \rightarrow$ **Scaling operation**

Example — $x(-4t)$

“Pasted image 20260720154740.png” could not be found.

Since both a negative sign *and* a coefficient are present, apply operations in order:

1. **Fold first** $\rightarrow$ obtain $x(-t)$
 2. **Then scale** $\rightarrow$ obtain $x(-4t)$
-

3. Scaling Operation

Key property: Like shifting, scaling **preserves shape and amplitude** — it compresses/expands the signal along the time axis (or in the folding+scaling combo, also mirrors it).

Quick Reference Summary

Operation	Trigger in $x(\cdot)$	Effect	Shape/Amplitude
Shifting	$t \pm a$ (addition/subtraction)	Moves signal left/right	Unchanged
Folding	$x(-t)$ (negative t)	Mirrors signal about y-axis	Unchanged
Scaling	$x(kt)$ (coefficient on t)	Compresses/expands signal	Unchanged

Combined operation order (e.g. $x(-kt)$): Fold $\rightarrow$ Scale